

DILAN QUINTERO ESPINOSA

Data Engineer | Azure Data Engineering | Industrial Automation

Cartagena, Colombia | Open to Remote Roles | +57 320 640 6958 | dylanquintero11@gmail.com

LinkedIn: [linkedin.com/in/dylan-quintero-43832a29b](https://www.linkedin.com/in/dylan-quintero-43832a29b) | GitHub: github.com/DylanQ94 | Portfolio: dylang94.github.io/Dylan_Portfolio

PROFESSIONAL SUMMARY

Mechatronics and Electrical Engineer transitioning from industrial automation and process engineering into Data Engineering. Hands-on experience with Python, SQL, Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, PySpark, Delta Lake, and Azure Synapse Serverless SQL through end-to-end Azure portfolio projects. Industrial background in PLC programming, HMI/SCADA development, process control, operational KPI analysis, root-cause analysis, and automated reporting. Strong ability to connect production environments, process data, and business requirements into reliable data solutions.

TECHNICAL SKILLS

Cloud & Data Platforms: Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, Azure Synapse Serverless SQL, Delta Lake

Programming & Analytics: Python, SQL, PySpark, pandas, matplotlib, data analysis, automated reporting

Data Engineering: ETL/ELT pipelines, Medallion Architecture, Lakehouse Design, batch processing, data validation, dimensional modeling, star schema

Databases & Formats: SQL Server 2012/2022, Delta tables, Parquet, CSV, JSON

Tools & Industrial Systems: Git, GitHub, Power BI, FactoryTalk View, FactoryTalk Transaction Manager, PLC, HMI, SCADA

Languages: Spanish: Native | English: Intermediate (B1)

PROFESSIONAL EXPERIENCE

Self-Employed Automation Engineer | Independent | Cartagena, Colombia | 2025 - Present

- Provide automation services for industrial environments while dedicating focused time to a professional transition into Data Engineering.
- Developed PLC programming for electrical power transfer control between plant supply and the utility grid.
- Built PLC and HMI logic for plastic roll drilling and cutting machinery, supporting machine control and operator interaction.
- Developed an automated reporting system using Python to extract data from a SQL Server 2012 database and generate process reports with pandas and matplotlib.
- Currently working on a dosing and mixing plant project, responsible for PLC programming, SCADA design for process control, and SQL Server 2022 database design using FactoryTalk View and FactoryTalk Transaction Manager logs.

Process Engineer | Ajover - Darnel | Cartagena, Colombia | 2023 - 2025

- Monitored and optimized six blown PVC film extrusion lines by analyzing operational data, production KPIs, and process behavior to identify improvement opportunities.
- Investigated availability, performance, and quality issues, supporting root-cause analysis and corrective action plans with maintenance and quality teams.
- Designed and evaluated automation and technical improvement initiatives aimed at increasing efficiency, process stability, and operational performance.
- Used production data and internal monitoring systems to support decision-making, troubleshooting, and continuous improvement in manufacturing operations.

Engineering Intern | Ajover - Darnel | Cartagena, Colombia | 2023

- Supported industrial automation projects in the Film plant, contributing to PLC logic development, PID control systems, HMI development, and electrical control panel analysis.
- Assisted with operational data entry and analysis in Excel, helping document production behavior and support engineering decisions.

SELECTED DATA ENGINEERING PROJECTS

Productivity Monitor Lakehouse on Azure | Portfolio Project | Remote | 2026

- Designed an Azure data engineering solution using ADLS Gen2, Azure Data Factory, Azure Databricks, PySpark, Delta Lake, and Synapse Serverless SQL to process productivity data from daily CSV/JSON files.
- Built Bronze, Silver, and Gold layers with dimensional modeling, fact tables, SQL views, and analytics-ready datasets for reporting consumption.

IoT Berkeley Lab Analytics Platform | Portfolio Project | Remote | 2026

- Implemented a lakehouse architecture for IoT sensor data using Azure Data Factory, ADLS Gen2, Azure Databricks, PySpark, Delta Lake, and Synapse Serverless SQL.
- Processed sensor readings and metadata through curated layers to support analytical querying and future dashboard consumption.

Azure Supply Chain Logistics Optimization Platform | Portfolio Project | Remote | 2026

- Developed an Azure-based analytical platform using ADLS Gen2, Databricks, Delta Lake, and Synapse Serverless SQL to prepare logistics and supply chain datasets for decision-making.
- Modeled curated Gold-layer tables and serverless SQL views to expose clean, queryable data for reporting and analysis.

EDUCATION

B.S. in Mechatronics Engineering | Universidad Tecnologica de Bolivar | Cartagena, Colombia | 2018 - 2023

B.S. in Electrical Engineering | Universidad Tecnologica de Bolivar | Cartagena, Colombia | 2018 - 2023

ADDITIONAL TRAINING

- Fundamentals of Project Planning and Management - University of Virginia, 2020
- Industrial Automation Seminar - Universidad Tecnologica de Bolivar, 2022